Noble gases

The **noble gases** are a special group of elements that exist as single atoms. They are very **stable** and rarely react with other elements. Electron shells each hold a different number of electrons. The noble gases have a full outer shell.

☞ **1** Use the information below to help fill in the table. Keep in mind that a shell does not need to be completely filled before electrons begin to fill the next level.

Electron shells

1	2	3	4
holds 2 electrons	holds 8 electrons	holds 8 or 18 electrons	holds 18 or 32 electrons

		Number of electrons in each shell				
Element	**Atomic number***	**1**	**2**	**3**	**4**	**5**
helium	2	2	-	-	-	-
neon	10					
argon	18					
krypton	36					

*Atomic number = number of electrons

When elements form a **compound** they fill their outer shell to become stable. Elements can react with each other to form compounds by giving, receiving or sharing electrons.

2 Can you explain why noble gases do not form compounds? _____

_____

3 Sodium has one electron in its outer shell. What do you think sodium does with this electron when it forms a compound? _____

4 Chlorine has 7 electrons in its outer shell. What do you think chlorine needs to do to fill its outer shell when it forms a compound? _____

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